PROJECT DURATION

June 1, 2026 – August 15, 2026

TEAM MEMBERS – 4

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Closed-Loop Delta Robot for Automated Inspection and Sorting

自動検査・仕分けのためのクローズドループ型デルタロボット

Problem Statement / 問題提起

Slow & Inconsistent Manual Sorting → **No Closed-Loop Verification** → **Silent Defects & Mis-Sorts** → **Costly Commercial Delta Cells** → **Open-Loop Positioning Errors**

Objectives (もくてき)

→ **Closed-Loop Pick & Place**
Servo + joint feedback

→ **Vision-Guided Classification**
Camera + Arduino

→ **Automated Sorting to Bins**
Actuated gripper

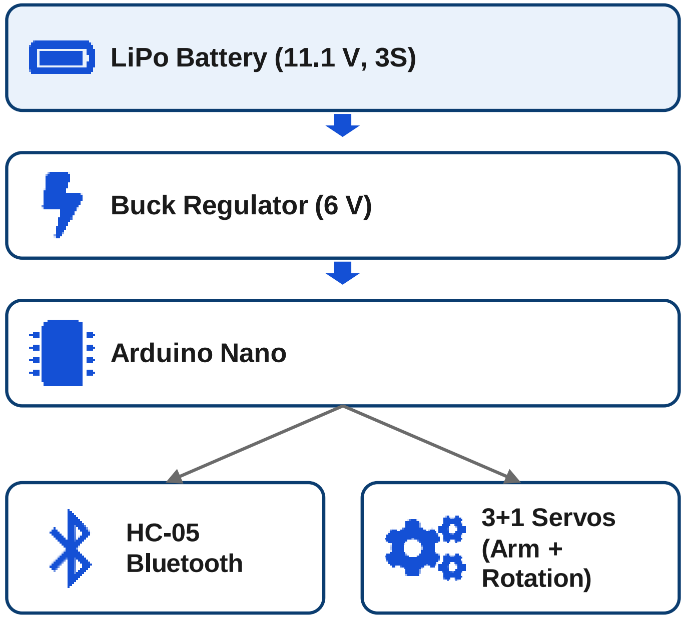
Prototype

Servo-driven parallel-link mechanism

SYSTEM ARCHITECTURE & METHODOLOGY

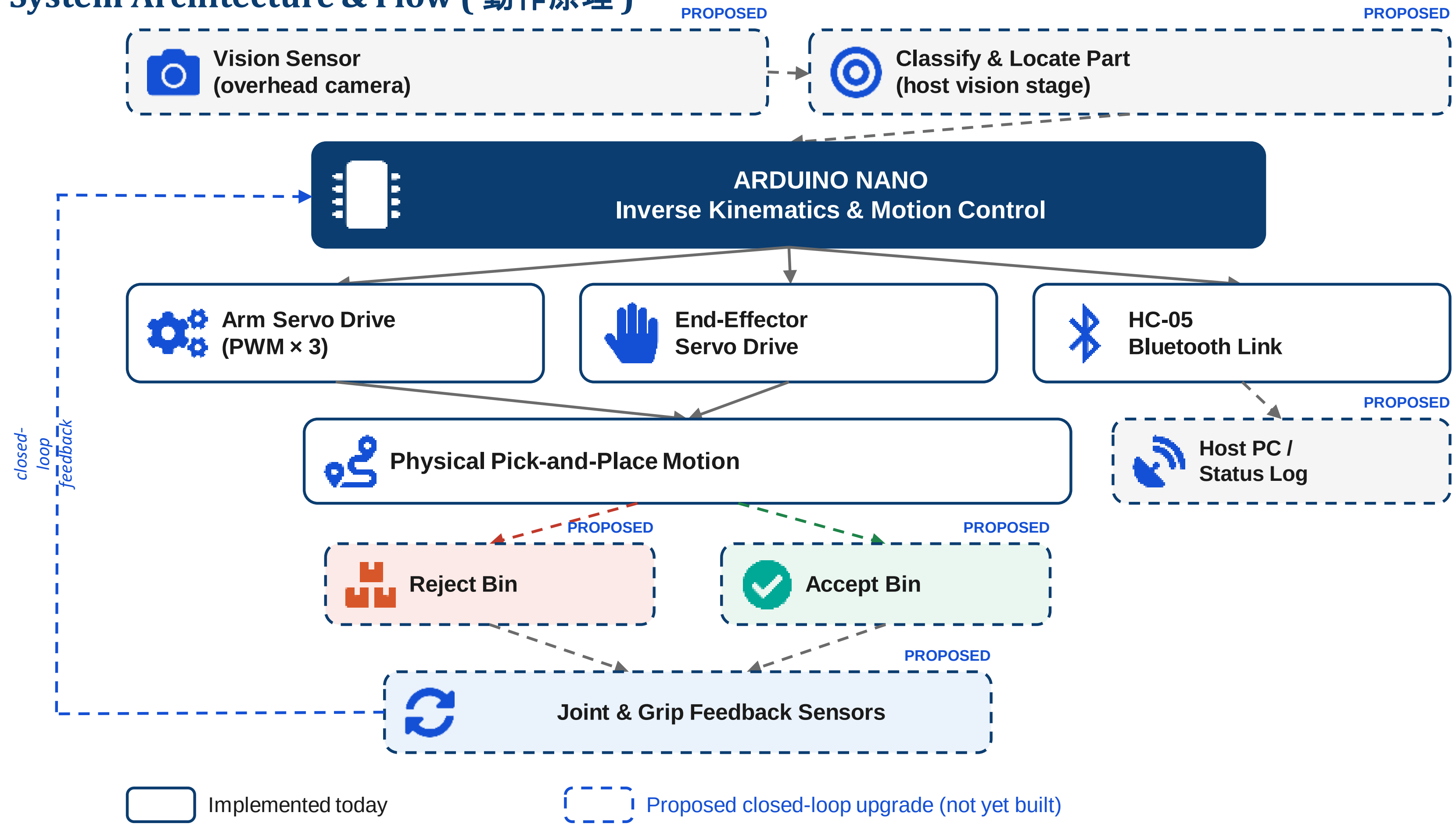
システム構成と手法

Hardware Architecture



Arduino Nano computes inverse kinematics and drives every servo's PWM channel.

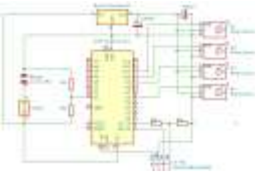
System Architecture & Flow (動作原理)



PROJECT OUTCOMES

Fully functioning, closed-loop-ready delta manipulator

Coordinated 3-DOF motion plus independent end-effector rotation



Syringe vacuum gripper tested to ~300 g lift, 24+ hr hold

FUTURE SCOPE

Joint Encoders for Closed-Loop Position Feedback

Pick-Confirmation Sensing for the Vacuum Gripper

Overhead Camera & Lightweight Part Classifier

Repeatability Benchmarking vs. Payload & Cycle Rate

PERFORMANCE CALCULATIONS

Metric	Basis / Calculation	Result
Battery energy capacity	11.1 V × 1.0 Ah (3S 1000 mAh pack)	11.1 Wh
Maximum discharge current	1.0 Ah × 20C rating	20 A
Servo-rail power budget	6 V × ~5 A rated regulator output	≈ 30 W
Est. battery-side input current	30 W ÷ 11.1 V ÷ assumed 85% efficiency	≈ 3.2 A
Leg tube mass (each)	$\pi/4 \times (16^2 - 14^2)$ mm² × 333 mm × 2.70 g/cm³	≈ 42 g
Total leg mass (3 legs)	≈ 42 g × 3	≈ 127 g
Vacuum gripper measured lift/hold	Padlock lift test / metal weight hold time	~300 g / 24+ hrs

Thank You

ありがとうございます

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